Prism Emission Calculation Methodology

Overview

This document describes the methodology used by Prism to estimate the operational emissions associated with AI usage during software development.

Prism is designed to provide visibility into AI-related emissions at the point where developers make decisions. The estimates generated by Prism are intended to support awareness, discussion, and informed decision-making.

Prism does not provide exact carbon measurements.

AI providers do not currently publish standardised, model-level energy consumption or emissions data for every request. As a result, all calculations rely on publicly available research, proxy data, and assumptions. The results should therefore be interpreted as directional estimates rather than precise measurements.

What Prism Measures

Prism estimates operational emissions associated with AI inference during software development.

This includes:

- AI-assisted development activities, such as coding assistants and chat-based development tools
- AI requests generated by applications during development, testing, and experimentation

Prism focuses on operational emissions because they are the emissions most directly influenced by developers through their day-to-day use of AI systems.

What Prism Does Not Measure

Prism does not currently account for:

- Model training emissions
- Hardware manufacturing emissions
- Data centre construction and infrastructure emissions
- Water consumption
- End-user production usage after deployment

- Wider system-level rebound effects

These factors may represent a significant portion of the overall environmental impact of AI systems but are currently outside the scope of Prism.

Methodological Approach

The Prism methodology draws on principles from the Green Software Foundation's Software Carbon Intensity specification.

The SCI specification is commonly expressed as:

```None
SCI = ((E × I) + M) ÷ R
```

Where:

```None
E = energy consumed by the software system

I = carbon intensity of the energy source

M = embodied emissions from hardware

R = functional unit
```

Prism adapts this structure for AI inference during software development.

In Prism:

```None
Estimated emissions = token activity × model-specific energy factor × carbon intensity factor
```

At a high level, the process consists of three stages:

1. Measure AI Activity

Prism records the number of input and output tokens associated with AI interactions.

Tokens provide a consistent way of estimating the amount of computation required to process a request.

2. Estimate Energy Consumption

Token counts are converted into estimated energy consumption using model-specific benchmark data derived from publicly available research.

Where model-specific data is unavailable, the closest available model or category is used as a proxy.

3. Estimate Operational Emissions

Estimated energy consumption is multiplied by a carbon intensity factor to estimate operational emissions.

Results are reported in grams of carbon dioxide equivalent (gCO_2e), a standard unit used to compare greenhouse gas emissions based on their global warming potential.

Carbon Intensity Assumptions

A significant challenge in estimating AI emissions is the lack of visibility into the physical location of the data centres processing requests.

Carbon intensity varies substantially between regions due to differences in electricity generation.

Because the location of AI inference workloads is generally unknown, Prism applies a global average grid carbon intensity value of 471 gCO_2e/kWh.

This approach provides a consistent basis for estimation while acknowledging that actual emissions may be higher or lower depending on the location and energy mix of the infrastructure used.

Data Sources

Prism uses publicly available research and emissions datasets to estimate model-level energy consumption.

The primary source is:

Jegham et al. (2025)
How Hungry is AI? Benchmarking Energy, Water, and Carbon Footprint of LLM Inference

Supporting benchmark data has also been sourced from publicly available industry tools and datasets where equivalent academic data was unavailable.

As new research becomes available, model conversion factors may be updated to improve accuracy and coverage.

Model Coverage

Prism supports a growing range of AI models.

Where published benchmark data exists for a specific model, that data is used directly.

Where benchmark data is unavailable, Prism uses the closest available model as a proxy.

Proxy models are selected based on available evidence regarding model architecture, capability, and expected computational requirements.

Estimation Method

Published benchmark data is not available for every possible prompt size or model interaction.

To address this limitation, Prism applies a tiered estimation approach based on the closest available benchmark data.

Where model-specific benchmark coverage is limited, estimates may overstate or understate actual emissions.

This approach was chosen to prioritise transparency and consistency while avoiding unsupported assumptions about model behaviour beyond the available evidence.

As additional benchmark data becomes available, the estimation methodology may be refined.

Sources of Uncertainty

Several factors contribute to uncertainty in AI emissions estimation.

These include:

- Limited transparency from AI providers
- Variations in model infrastructure
- Unknown data centre locations
- Differences in hardware efficiency
- Evolving model architectures
- Incomplete public benchmark data

As a result, emissions estimates should be interpreted as indicative rather than definitive.

The purpose of Prism is not to provide carbon accounting-grade accuracy, but to provide a practical signal where little visibility currently exists.

Interpretation Guidance

Prism estimates are most useful when used to:

- Compare patterns within a project
- Identify unusually resource-intensive interactions
- Track trends over time
- Support conversations about responsible AI usage

Prism estimates are not intended to:

- Demonstrate regulatory compliance
- Produce formal greenhouse gas inventories
- Compare organisations
- Verify sustainability claims

Future Improvements

The Prism methodology will continue to evolve as better data becomes available.

Future areas of exploration include:

- Improved model-specific benchmark coverage
- Region-aware carbon intensity estimates
- Water consumption estimates
- Enhanced support for image and multimodal models
- Improved uncertainty modelling
- Additional public research integrations

Disclaimer

Prism provides directional estimates intended to support awareness and decision-making.

The methodology relies on the best publicly available research and data available at the time of development. Results should not be interpreted as exact measurements of environmental impact.

As the availability and quality of AI emissions data improves, Prism's methodology and estimates may change accordingly.